Northwestern

Andre Shportko <andreshportko2026@u.northwestern.edu>

AI & Metacognition paper progress <Shportko & Agnoli>

13 messages

Andre Shportko <andreshportko2026@u.northwestern.edu> Fri, Apr 10, 2026 at 3:16 PM
To: Jessica Ruth Hullman <jhullman@northwestern.edu>
Cc: Sam Agnoli <SamAgnoli@u.northwestern.edu>

Dear Professor,

Thank you for providing feedback to our work earlier this spring. We are now ready to resume the work, and before we do so, I would like to offer an updated methodology as well as to request feedback on it.

Methodology:

- Establish baselines (for example, explore if prompt-engineering alone can improve the intra-model accuracy-confidence agreement). For the baseline, I recommend doing 0-shot, 1-shot, and 5-shot settings in order to simulate "training".
- Since we can't test 166 models, there are few things we can do. First option is what I think is more methodologically sound, second is what we did for the seminar, but the comparisons inferred from the latter are less faithful to human subjects:
  1) Simulate 166 personas on the same model based on demographics data provided in the dataset, including age, gender, race/ethnicity, education, and political affiliation variables. If we do so, we can more faithfully evaluate how steering of the personas population brings the model closer to the human population in the 80-dimensional [1 dimension per question] space of confidence scores.
  2) Rerun the same model under temperature = 0.5 for 166 times. In this case, we can't perform comparisons with human populations in a one-to-one manner as faithfully as we could with option one.
- For confidence self-report, I strongly prefer to use the logit-based approach as outlined in the paper, because it provides the most unbiased self-reflection that the model can offer in a way that is both parsable and error-free.
- I recommend keeping the loss function as it is. If it is suboptimal, I am still looking for a metric that balances between metacognition alignment and the preservation of general performance (i.e, lobotomy detection). Could you please recommend or remind us of your suggestions?
- I am also open to your suggestions for additional metrics that would strengthen the manuscript.
- We will merge E1 and E8, and will incorporate your other feedback on the experimental sequence.

Regarding the deadlines, there are few venues we are considering:

1) ICML 2026 mech interp https://openreview.net/group?id=ICML.cc/2026/Workshop/Mech_Interp#tab-recent-activity
Deadline: May 09, 2026, 6:59:00 AM
2) 2026 AI in Social Science Conference https://bfi.uchicago.edu/events/event/2026-ai-in-social-science-conference/
Deadline: May 01
They say it is peer-reviewed, but I don't understand where it will be published
3) ICML 2026: AI and philosophy (they mention cognition and explainability) https://sites.google.com/view/philmlicml26/home
11th May: Paper submission deadline

Please suggest to us if there is a more appropriate venue for us to be platformed at, even if this venue is not listed above.

We are excited to co-author this work with you, and we believe that this work can be driven to completion this month. Once we receive your feedback, we will be ready to redo the experimental pipeline, following a new protocol.

Best,

Andre

Jessica Ruth Hullman <jhullman@northwestern.edu> Fri, Apr 10, 2026 at 3:55 PM
To: Andre Shportko <AndreShportko2026@u.northwestern.edu>
Cc: Sam Agnoli <SamAgnoli@u.northwestern.edu>

Thanks! I'll aim to get you feedback this weekend, but one other venue that would be easy to submit to/attend is the Communication & Intelligence workshop at UChicago poster call - deadline is April 15: https://ci.cs.uchicago.edu/symposium/

Jessica

Jessica Hullman
Ginni Rometty Professor of Computer Science
Faculty Fellow, Institute for Policy Research
Northwestern University
[Quoted text hidden]

Jessica Ruth Hullman <jhullman@northwestern.edu> Sun, Apr 12, 2026 at 7:16 PM
To: Andre Shportko <AndreShportko2026@u.northwestern.edu>
Cc: Sam Agnoli <SamAgnoli@u.northwestern.edu>

Here's my feedback:

- Establish baselines (for example, explore if prompt-engineering alone can improve the intra-model accuracy-confidence agreement). For the baseline, I recommend doing 0-shot, 1-shot, and 5-shot settings in order to simulate "training".

Yes, prompting baselines make sense, but would you only look at accuracy / confidence agreement, or compare the different response distributions to human data?

- Since we can't test 166 models, there are few things we can do. First option is what I think is more methodologically sound, second is what we did for the seminar, but the comparisons inferred from the latter are less faithful to human subjects:

1) Simulate 166 personas on the same model based on demographics data provided in the dataset, including age, gender, race/ethnicity, education, and political affiliation variables. If we do so, we can more faithfully evaluate how steering of the personas population brings the model closer to the human population in the 80-dimensional [1 dimension per question] space of confidence scores.

2) Rerun the same model under temperature = 0.5 for 166 times. In this case, we can't perform comparisons with human populations in a one-to-one manner as faithfully as we could with option one.

I agree that first option makes more sense. If possible I'd simulate several samples of 166 participants (like at least 5) and report the mean plus uncertainty.

- For confidence self-report, I strongly prefer to use the logit-based approach as outlined in the paper, because it provides the most unbiased self-reflection that the model can offer in a way that is both parsable and error-free.

Sounds good.

- I recommend keeping the loss function as it is. If it is suboptimal, I am still looking for a metric that balances between metacognition alignment and the preservation of general performance (i.e, lobotomy detection). Could you please recommend or remind us of your suggestions?

I think the current loss function has a few problems: dividing baseline by steered accuracy makes the objective non-linear, the 50/50 weighting seems arbitrary, and ECE is known to be very sensitive to binning and sample size.

I think it could make sense to incorporate a proper scoring rule. Basically rather than 1 though 4, elicit the model's confidence as a probability. Then you can measure confidence calibration as Brier loss (relative to the ground truth in the text phase - 0 or 1 for whether model was right or wrong), and you could still compare to human confidence by doing a mapping from the human responses to a probability (there are a few ways to do that).

But first I need more info on why your current loss is penalizing going too far from baseline loss. What's the reason for that? (Why is baseline considered good?)

- I am also open to your suggestions for additional metrics that would strengthen the manuscript.

In general I think you want to look at how close you get to the human data in terms of response distributions themselves. I think you had this in the last version, but I recall more focus on the correlation between confidence and accuracy.

- We will merge E1 and E8, and will incorporate your other feedback on the experimental sequence.
Regarding the deadlines, there are few venues we are considering:

1) ICML 2026 mech interp https://openreview.net/group?id=ICML.cc/2026/Workshop/Mech_Interp#tab-recent-activity
Deadline: May 09, 2026, 6:59:00 AM

Seems like a solid choice

2) 2026 AI in Social Science Conference https://bfi.uchicago.edu/events/event/2026-ai-in-social-science-conference/
Deadline: May 01
They say it is peer-reviewed, but I don't understand where it will be published

Probably no publication - I'm guessing its just talks and maybe posters.

3) ICML 2026: AI and philosophy (they mention cognition and explainability)
https://sites.google.com/view/philmlicml26/home
11th May: Paper submission deadline

This could also be worth submitting to.

Jessica

Jessica Hullman
Ginni Rometty Professor of Computer Science
Faculty Fellow, Institute for Policy Research
Northwestern University

From: Andre Shportko <andreshportko2026@u.northwestern.edu>
Date: Friday, April 10, 2026 at 3:17 PM
To: Jessica Ruth Hullman <jhullman@northwestern.edu>
Cc: Sam Agnoli <SamAgnoli@u.northwestern.edu>
Subject: AI & Metacognition paper progress <Shportko & Agnoli>

[Quoted text hidden]

Andre Shportko <andreshportko2026@u.northwestern.edu> Mon, Apr 13, 2026 at 10:24 PM
To: Jessica Ruth Hullman <jhullman@northwestern.edu>
Cc: Sam Agnoli <SamAgnoli@u.northwestern.edu>

Thank you for the detailed feedback.

Baselines: we already compare the full response distributions (correct, lure, unsure, other) to human data. It's just now we can do it across the 0-shot, 1-shot, and 5-shot conditions.

Persona simulation: we will go with option 1 and simulate at least 5 independent samples of 166 persona-based

participants. Before running, I will try to project how expensive it will be, and then once estimated, I might recommend splitting the costs if it is too prohibitive in light of reasonable alternatives.

Loss function: the reason baseline accuracy appears in the denominator is as a "lobotomy guard" because we wanted to penalize steering configurations that improve calibration at the cost of destroying the model's ability to answer questions correctly. It is not that baseline performance is considered good in itself, but it's just to ensure steering doesn't degrade accuracy below the unsteered level.

That said, we will explore replacing ECE with Brier loss as the calibration component.

$$\text{Brier} = \frac{1}{N}\sum_{i=1}^{N}(\text{conf}_i - \text{correct}_i)^2$$

$$L = |\text{Brier}_{\text{model}} - \text{Brier}_{\text{human}}| + (1 - r_{\text{conf,acc}}) + \lambda \cdot \max(0,\ \text{acc_base} - \text{acc_steered})$$

Will this loss function address non-linearity and ECE issues with binning? For AI, I normalized logits that produce a value in [0,1], so I can compute the model Brier directly. For humans, I'd map the 1–4 scale to probabilities. Few options there:

1.Linear: p=(r−1)/3p = (r - 1)/3
2.Empirical: from the calibration split, compute the actual accuracy rate for participants who reported each confidence level (e.g., people who said "3" were correct 72% of the time -> map 3 to 0.72). I suggest rolling with option 2 b/c it is more empirically motivated.

The loss function is the most important part of this paper that advises our coefficient selection, so we will highly appreciate your corrections to the final Loss form. Once finalized, I can run it all this week by Wednesday (I will fly to the AI control conference this Thursday).

Metrics: We will add distributional comparisons (Wasserstein distance or KL divergence between model and human response category distributions), in addition to the confidence-accuracy correlation.

As you validated, we are planning to target the ICML Mech Interp workshop (May 9) as the primary submission and the AI & Philosophy workshop (May 11) as a secondary. We will keep you updated on progress.

Best,
Andre
[Quoted text hidden]

Jessica Ruth Hullman <jhullman@northwestern.edu> Tue, Apr 14, 2026 at 9:18 PM
To: Andre Shportko <AndreShportko2026@u.northwestern.edu>
Cc: Sam Agnoli <SamAgnoli@u.northwestern.edu>

Ok sounds good.

On loss function, Brier loss with the empirical option is closest to what I was thinking. But why do you still need the reference to baseline accuracy in the functional form? Can’t you drop that now that you have prior? The Brier score is looking at how well their confidence predicts the true state, so in a sense it takes care of the accuracy portion.
[Quoted text hidden]

Jessica Ruth Hullman <jhullman@northwestern.edu> Tue, Apr 14, 2026 at 9:20 PM
To: Andre Shportko <AndreShportko2026@u.northwestern.edu>
Cc: Sam Agnoli <SamAgnoli@u.northwestern.edu>

One other note on loss function — whether you keep the final term or not, a good way to design these things is to make sure that the limiting behavior of each function makes sense. E.g., for each term, what is the value of it when the responses are as bad as possible versus as good as possible.

Jessica

Jessica Hullman
Ginni Rometty Professor of Computer Science
Faculty Fellow, Institute for Policy Research
Northwestern University
[Quoted text hidden]

Andre Shportko <andreshportko2026@u.northwestern.edu> Tue, Apr 14, 2026 at 10:29 PM
To: Jessica Ruth Hullman <jhullman@northwestern.edu>
Cc: Sam Agnoli <SamAgnoli@u.northwestern.edu>

I want a loss function that on one hand aligns metacognition but not at the cost of the accuracy of the answers. How would Brier score account for that?

Best,

Andre
[Quoted text hidden]

Jessica Ruth Hullman <jhullman@northwestern.edu> Wed, Apr 15, 2026 at 8:59 AM
To: Andre Shportko <AndreShportko2026@u.northwestern.edu>
Cc: Sam Agnoli <SamAgnoli@u.northwestern.edu>

Brier score accounts for correctness in the sense that if the model is accurate and gives high probabilities on the correct answer, its Brier score will be low, and high if vice versa. But the less the probabilities you elicit represent the model's 'true' decision probabilities, the less you learn about accuracy from it, and even when they are the model's true probabilities, it's not complete as a measure of accuracy in that you still don't know the model's decision boundary. So it's related but not synonymous.

If you want a separate term for accuracy, shouldn't that be relative to human accuracy to keep things consistent?
[Quoted text hidden]

Andre Shportko <andreshportko2026@u.northwestern.edu> Wed, Apr 15, 2026 at 11:59 AM
To: Jessica Ruth Hullman <jhullman@northwestern.edu>
Cc: Sam Agnoli <SamAgnoli@u.northwestern.edu>

Yes that makes sense!

So let's say my loss function is a linear combination of three terms:

L= |Brier_model−Brier_human|+alpha*|rho_model−rho_human|+beta*max(0, acc_human−acc_steered)

1) delta Brier is how far the model's calibration is from human calibration
2) delta pro is how far the model's metacognitive sensitivity is from human sensitivity
3) max(0, acc_human − acc_steered) is penalty if steering makes the model dumber than humans

Now everything is baselined against humans. Is it better?

Best,

Andre

[Quoted text hidden]

2 attachments

$$\mathcal{L} = |\text{Brier}_{\text{model}} - \text{Brier}_{\text{human}}| + (1 - r_{\text{conf,acc}}) + \lambda \cdot \max(0,\ \text{acc_base} - \text{acc_steered})$$

image.png
14K

$$\text{Brier} = \frac{1}{N}\sum_{i=1}^{N}(\text{conf}_i - \text{correct}_i)^2$$

image.png
8K

Jessica Ruth Hullman <jhullman@northwestern.edu> Wed, Apr 15, 2026 at 12:20 PM
To: Andre Shportko <AndreShportko2026@u.northwestern.edu>
Cc: Sam Agnoli <SamAgnoli@u.northwestern.edu>

I assume the last term has a different form because you don't want to penalize the model being more accurate than humans. But I could see going either way on that — superhuman performance in LLM simulations can also be a bad thing if you're using them to learn about humans. But I don't have a strong opinion either way.

[Quoted text hidden]

Andre Shportko <andreshportko2026@u.northwestern.edu> Wed, Apr 15, 2026 at 12:53 PM
To: Jessica Ruth Hullman <jhullman@northwestern.edu>
Cc: Sam Agnoli <SamAgnoli@u.northwestern.edu>

The last term only penalizes sub-human performance, and does nothing (+0) if superhuman. Is this ok given that accuracy rise side-effect isn't a factor of the steering coefficient selection?

Furthermore, I suggest beta = 10 and alpha = 1 to penalize accuracy degeneration by a magnitude more than metacognition misalignment.

Best,

Andre

[Quoted text hidden]

Jessica Ruth Hullman <jhullman@northwestern.edu> Wed, Apr 15, 2026 at 1:30 PM
To: Andre Shportko <AndreShportko2026@u.northwestern.edu>
Cc: Sam Agnoli <SamAgnoli@u.northwestern.edu>

It's probably fine, or you could try a few ways if you have time (or choose one and run an ablation later if there's time).

[Quoted text hidden]

Andre Shportko <andreshportko2026@u.northwestern.edu> Wed, Apr 15, 2026 at 1:39 PM
To: Jessica Ruth Hullman <jhullman@northwestern.edu>
Cc: Sam Agnoli <SamAgnoli@u.northwestern.edu>

Okay, we will work on it today

Best,

Andre

[Quoted text hidden]